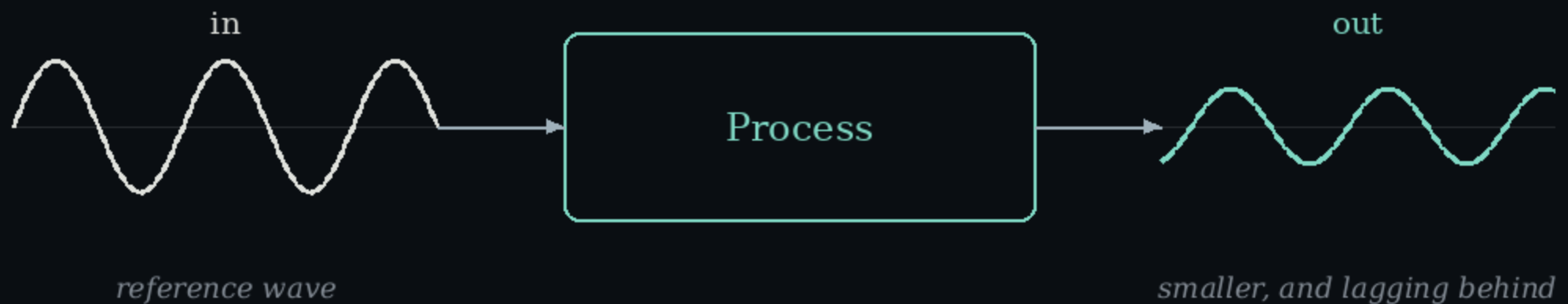




A QUICK ENGINEERING BREATH

THE SINE SWEEP, AT A GLANCE

Feed the process a sine wave. Watch what comes out the other side.



Raise the frequency ω . The lag grows and the output keeps shrinking — until the lag reaches 180° .

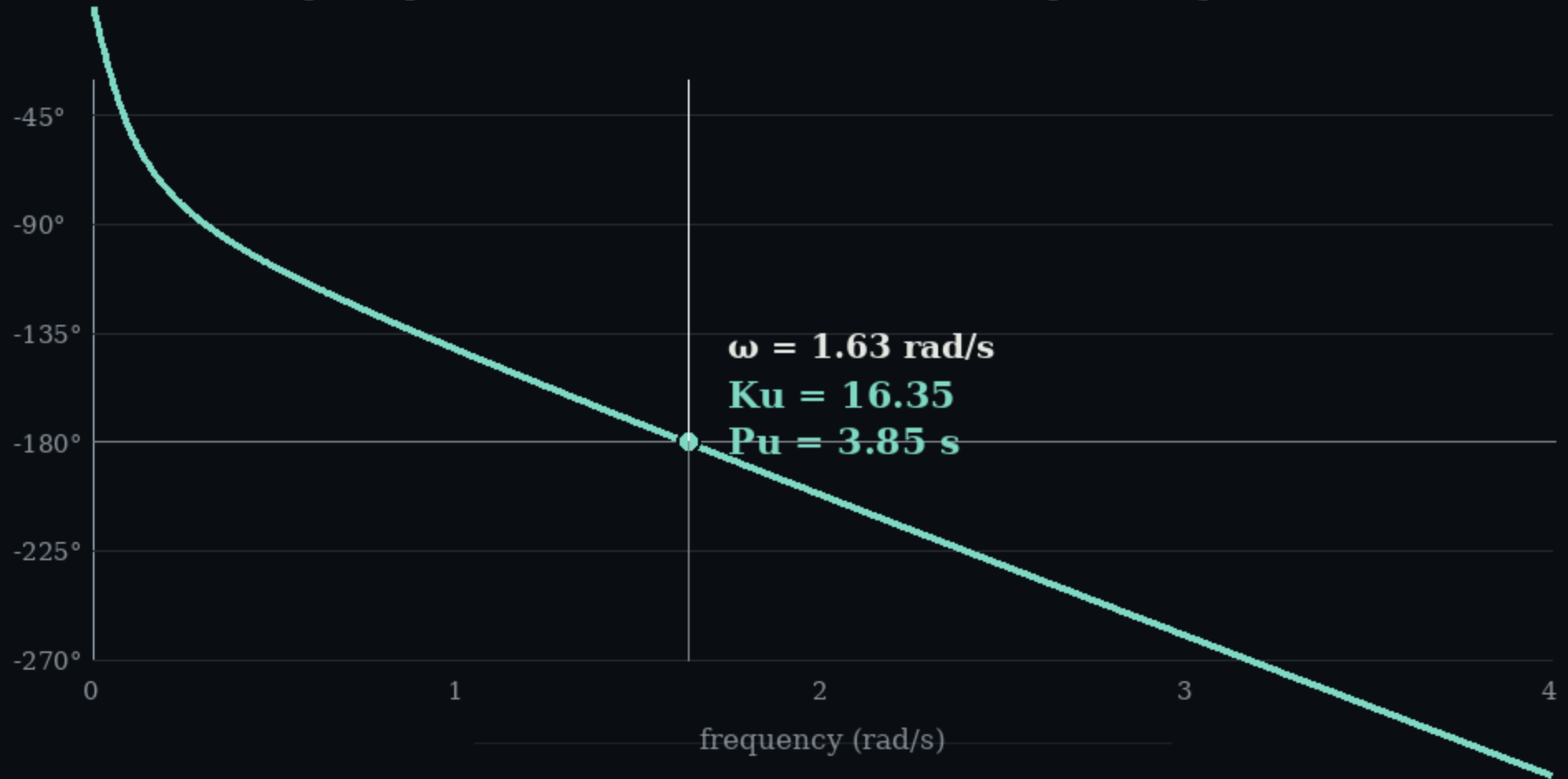
That crossing point is where Ku and Pu come from — next.



A QUICK ENGINEERING BREATH

WHERE K_u AND P_u ACTUALLY COME FROM

Sweep the process with a sine wave. Track the phase lag. Find -180° .



The same numbers behind last week's three tuning curves.



A QUICK ENGINEERING BREATH

TWO NUMBERS, THREE PERSONALITIES

$K_u = 16.35$

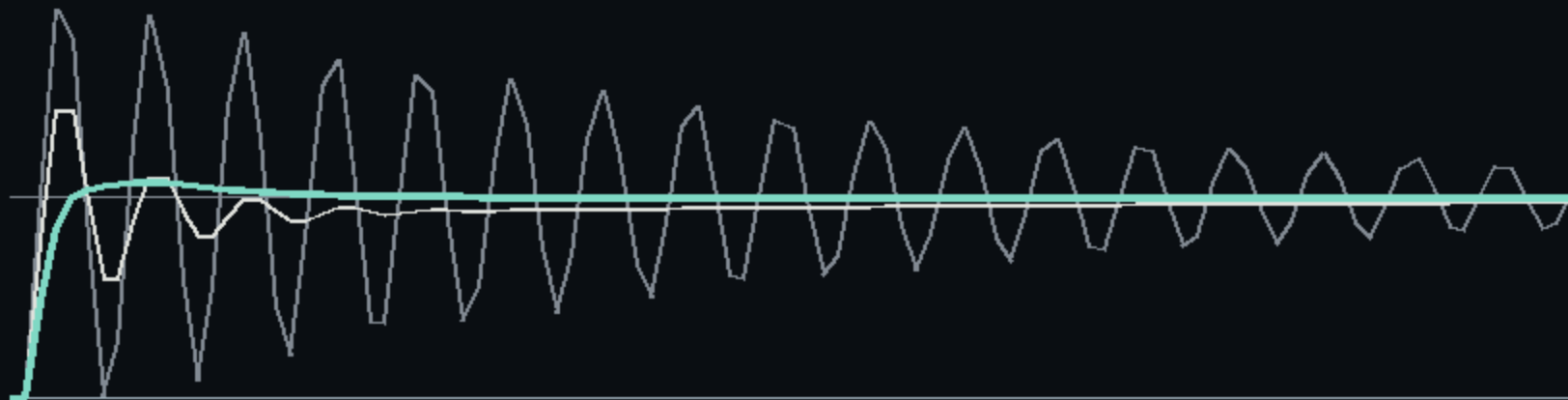
$P_u = 3.85 \text{ s}$

same two numbers — three different tuning rules

— Classic ZN
aggressive

— Tyreus-Luyben
balanced

— No-overshoot
conservative



Last week's post — now you've seen where it starts.